

A Biomolecular Sensing Platform with Floating Gate Ion-Sensitive Field Effect Transistors (FG-ISFETs)

Fan Xue¹, Yizhou Chen¹, Alana Yang¹, Sebastian Meier², Ernst Mueller², Ahmad Bahai^{1,2}, Joel Voldman¹

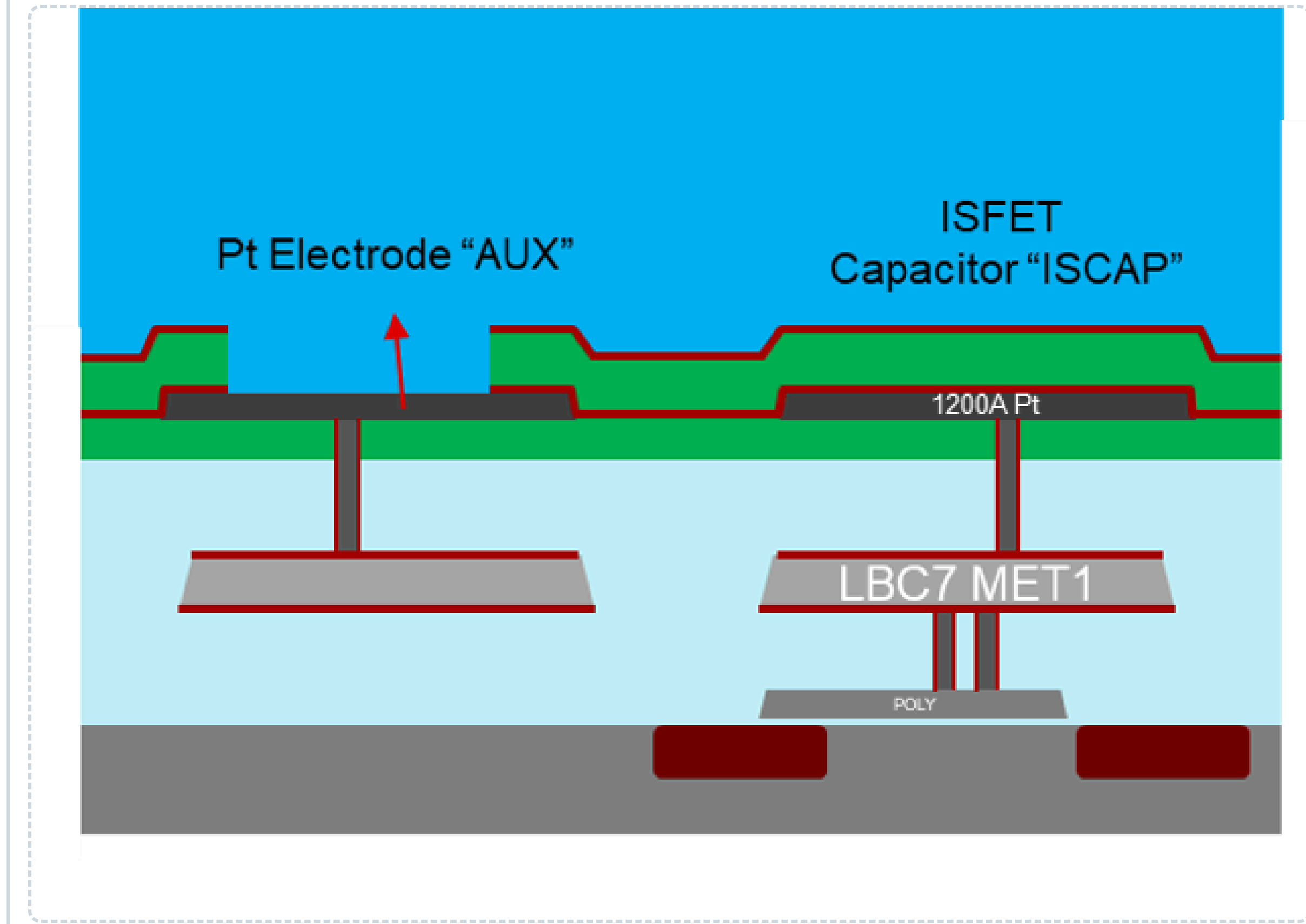
Motivation

Continuous biomarker monitoring could improve disease diagnosis, therapeutic management, and personalized healthcare. However, many existing biosensing approaches are limited by baseline drift, insufficient long-term stability, and difficulty integrating selective recognition layers with scalable electronics.

FG-ISFETs are attractive for this problem because they convert surface potential changes into electrical signals while preserving compatibility with semiconductor fabrication and integrated readout circuitry.

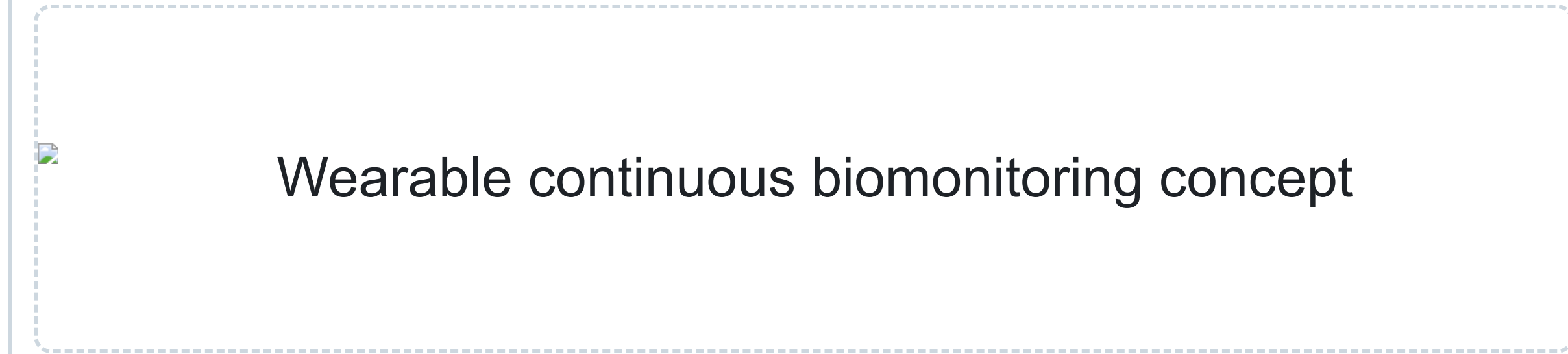
Overall goal: build a CMOS-compatible FG-ISFET platform for future aptamer-based continuous biomolecular sensing.

Device Platform



- Floating-gate architecture:** the sensing surface is capacitively coupled to the transistor channel.
- Electrical readout:** surface chemical changes are converted into measurable transistor output signals.
- CMOS compatibility:** scalable manufacturing, multiplexing, and on-chip electronics.
- Ta₂O₅ interface:** pH-sensitive oxide surface and substrate for silane-based functionalization.

Application Vision

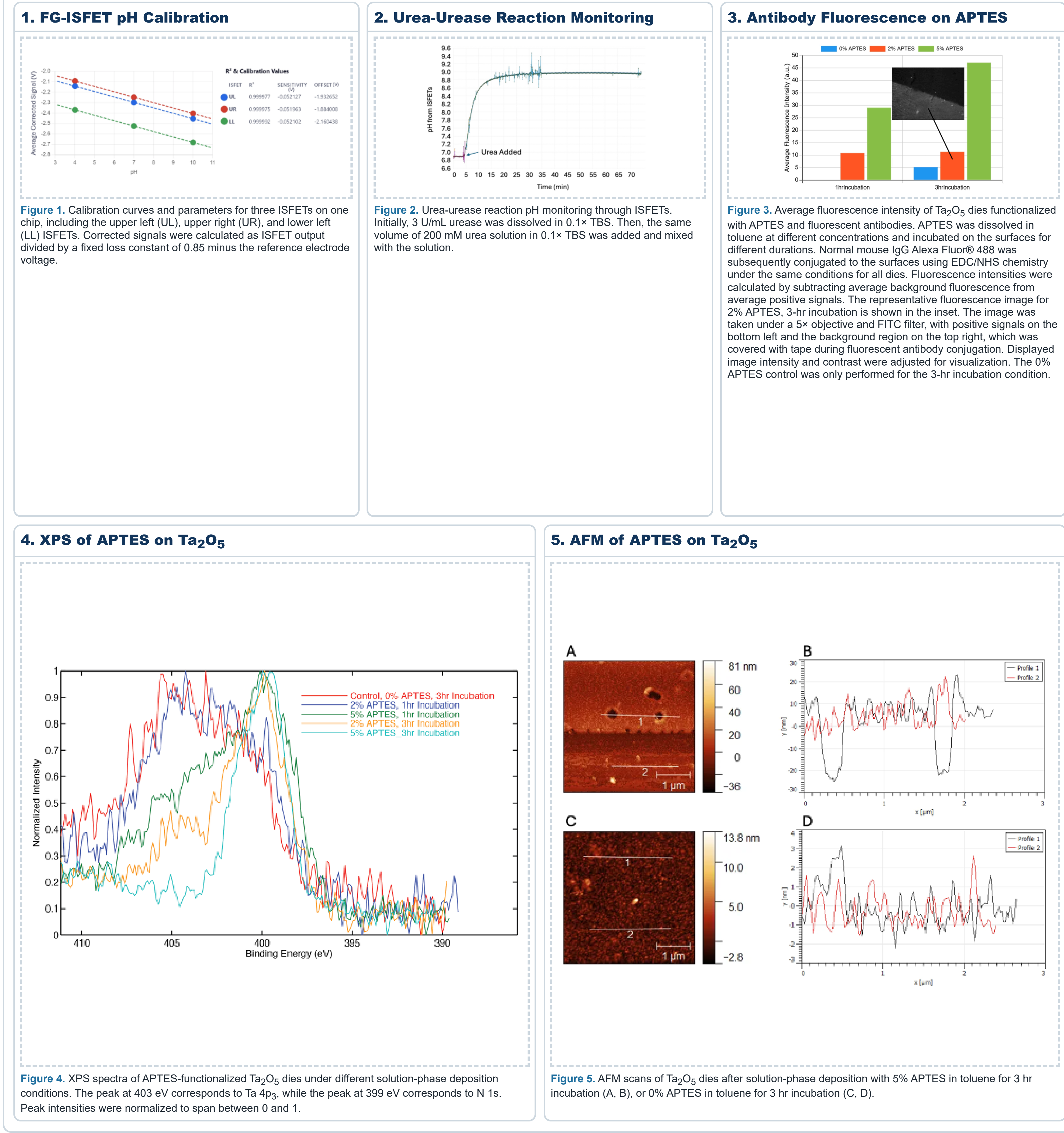


The long-term vision is to integrate the FG-ISFET sensing platform into a wearable or minimally invasive format for continuous biomarker monitoring in body fluids such as sweat or interstitial fluid.

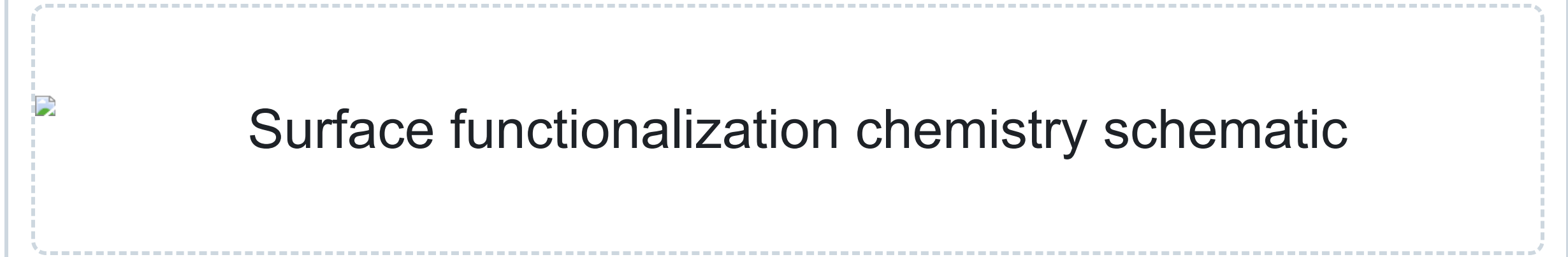
Molecular information from the sensing interface would be converted into time-resolved electrical signals, enabling longitudinal tracking of biomarker dynamics rather than isolated single-point measurements.

Wearable sensing	Continuous readout
Body-fluid monitoring	Personalized health

Results



Surface Functionalization Strategy



The sensing surface must connect the inorganic FG-ISFET interface to a biomolecular recognition layer. APTES introduces primary amine groups onto Ta₂O₅, enabling downstream coupling chemistry for aptamer immobilization.



The desired surface is thin, uniform, reproducible, and chemically accessible. Excessive multilayer formation can increase roughness, reduce reproducibility, and make electrical readout harder to interpret.

Future Steps

Surface Chemistry

- Optimize vapor-phase APTES deposition.
- Improve monolayer uniformity.
- Quantify surface amine availability.

Aptamer Integration

- Immobilize aptamers on functionalized Ta₂O₅.
- Control aptamer density and orientation.
- Preserve binding activity after surface attachment.

Target Detection

- Detect target-induced conformational changes.
- Translate binding into FG-ISFET signal shifts.
- Evaluate response in relevant buffer conditions.

Sensor Performance

- Investigate long-term stability.
- Evaluate selectivity and sensitivity.
- Measure dose response and detection limits.

Future Regeneration Strategy Comparison
Future work will compare passive dissociation, chemical regeneration, and electrochemical regeneration strategies for restoring sensor response after target binding. Regeneration is presented here as a future direction rather than a demonstrated result.

Conclusions

- FG-ISFETs provide a scalable electrical platform for biomolecular sensing.
- pH calibration and urease reaction measurements validate chemical signal readout.
- Antibody fluorescence, XPS, and AFM together characterize APTES-modified Ta₂O₅ surfaces.
- AFM suggests solution-phase APTES can form non-uniform layers, motivating vapor-phase deposition.
- Next steps focus on aptamer immobilization, conformational-change detection, and long-term sensor evaluation.

